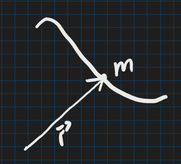
Quinta clase

Friday, August 21, 2026 10:41 AM

1) $S = 3N - K$ donde K son los vínculos del sistema, N las coordenadas

$V(r) = \frac{1}{2} r^2 \left(\frac{1}{2} K_1 r^2 - K_2\right)$



$v^2 = v_x^2 + v_y^2 + v_t^2$

$v = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\dot{\varphi}\,\mathrm{sen}\,\theta\,\hat{\varphi}$

$v^2 = \dot{r}^2 + r^2\dot{\theta}^2 + r^2\dot{\varphi}^2\,\mathrm{sen}^2\theta$

$\mathcal{L}(r, \dot{r}, \dot{\theta}, \dot{\varphi})$

$\frac{\partial \mathcal{L}}{\partial r} = m\dot{\theta}^2 + m r \dot{\varphi}^2 \,\mathrm{sen}\,\theta + r K_2^2 - r^3 K_1^2$

$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{1}{2} m r^2 \,\mathrm{sen}\,\theta\,\dot{\varphi}^2 \cdot \cos\theta$

$\frac{\partial \mathcal{L}}{\partial \theta} = m r^2 \,\mathrm{sen}\,\theta \cos\theta\,\dot{\varphi}^2$

$\frac{\partial \mathcal{L}}{\partial \varphi} = 0$

$\mathcal{L} = T - V$

$\mathcal{L} = \frac{1}{2} m v^2 - \frac{1}{2} r^2 \left(\frac{1}{2} K_1 r^2 - K_2\right)$

$= \frac{1}{2}\left(m v^2 - r^2\left(\frac{1}{2} K_1 r^2 - K_2\right)\right)$

$= \frac{1}{2}\left(m v^2 + r^2\left(K_2^2 - \frac{1}{2} K_1 r^2\right)\right)$

$= \frac{1}{2}\left(m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\dot{\varphi}^2\,\mathrm{sen}^2\theta\right) + r^2\left(K_2^2 - \frac{1}{2} K_1 r^2\right)\right)$

$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0$

$= \frac{1}{2} m\dot{r}^2 + \frac{1}{2} m r^2\dot{\theta}^2 + \frac{1}{2} m r^2 \dot{\varphi}^2 \,\mathrm{sen}^2\theta + \frac{1}{2} r^2 K_2^2 - \frac{1}{4} r^4 K_1^2$

$\frac{d}{dt}(r^2\dot{\theta}) = 2r \cdot \dot{r}\dot{\theta} + r^2\ddot{\theta}$

$\frac{\partial \mathcal{L}}{\partial \dot{r}} = m\dot{r}$ $\quad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{r}}\right) = m\ddot{r}$

$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m r^2\dot{\theta}$ $\quad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}\right) = 2m r \cdot \dot{r}\dot{\theta} + r^2\ddot{\theta}$

$\frac{\partial \mathcal{L}}{\partial \dot{\varphi}} = m r^2 \dot{\varphi}\,\mathrm{sen}^2\theta$ $\quad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\varphi}}\right) = 2m r\dot{r}\dot{\varphi}\,\mathrm{sen}^2\theta + m r^2\ddot{\varphi}\,\mathrm{sen}^2\theta + m r^2\dot{\varphi}\,\mathrm{sen}\theta \cdot \cos\theta \cdot \dot{\theta}$

5. El Lagrangiano de una partícula de masa m es

$$\mathcal{L} = \frac{m^2\dot{x}^4}{12} + m\dot{x}^2 f(x) - f^2(x)$$

donde $f(x)$ es una función diferenciable de x. Encuentre la ecuación de movimiento.

$\mathcal{L} = \frac{m^2\dot{x}^4}{12} + m\dot{x}^2 f(x) - f^2(x)$

$\mathcal{L}\{x, \dot{x}, y, \dot{y}\}$

$\frac{\partial \mathcal{L}}{\partial x} = m\dot{x}^2 f'(x) - 2f(x) \cdot f'(x)$

$\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{1}{3} m^2\dot{x}^3 + 2m\dot{x} f(x)$

$m^2 \cdot \dot{x}^2 \cdot \ddot{x} + 2m\ddot{x} f(x) + 2m\dot{x}^2 f'(x) - m^2\dot{x} f'(x) - 2f(x) \cdot f'(x) = 0$

$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial q_j}\right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0$

$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) = m^2 \cdot \dot{x}^2 \cdot \ddot{x} + 2m(\ddot{x} f(x) + \dot{x} f'(x) \cdot \dot{x})$

$\frac{d}{dt}(2m\dot{x} f(x)) = 2m\frac{d}{dt}(\dot{x} f(x)) = (\ddot{x} f(x) + \dot{x}\frac{d}{dt} f(x))$

$\frac{d}{dt}(f(x(t)) = \frac{df}{dx} \cdot \frac{dx}{dt} = f'(x) \cdot \dot{x}$

$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) = m^2 \cdot \dot{x}^2 \cdot \ddot{x} + 2m\ddot{x} f(x) + 2m\dot{x}^2 f'(x))$

$\frac{d}{dt}(m r^2\dot{\varphi}\,\mathrm{sen}^2\theta) = m\frac{d}{dt}(r^2\dot{\varphi}\,\mathrm{sen}^2\theta) = m(2r \cdot \dot{r}\dot{\varphi}\,\mathrm{sen}^2\theta + r^2\ddot{\varphi}\,\mathrm{sen}^2\theta + r^2\dot{\varphi}^2 \,\mathrm{sen}\theta \cdot \cos\theta \cdot \dot{\theta})$

4. Un cuerpo se deja caer desde una altura h y alcanza el suelo en un tiempo T. La ecuación de movimiento concebiblemente podría tener cualquiera de las formas

$$y = h - g_1 t, \quad y = h - \frac{1}{2} g_2 t^2, \quad y = h - \frac{1}{4} g_3 t^3$$

donde g_1, g_2, g_3 son constantes apropiadas. Demuestre que la forma correcta es aquella que produce el valor mínimo de la acción.

$\mathcal{L}\{q_i, \dot{q}_i, t\} = T - V$

$\mathcal{L} = \frac{1}{2} m\dot{y}^2 - mgy$

$V = mgh$

$T = \frac{1}{2} m v^2$

$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} = 0$

$\frac{\partial \mathcal{L}}{\partial y} = -mg$

$\frac{\partial \mathcal{L}}{\partial \dot{y}} = m\dot{y}$

$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{y}}\right) = m\ddot{y}$

$x(t) = -g\frac{t^2}{2} + h$

$m\ddot{y} - (-mg) = 0$

$m\ddot{y} + mg = 0$

$\ddot{y} + g = 0 \qquad \ddot{y} = -g$

$\ddot{y} = \dot{v}$

$\dot{v} = -g$

$\frac{dv}{dt} = -g \qquad v(0) = 0$

$dv = -g\,dt$

$v = -gt + c$

$0 = c$

$v = -gt$

$\frac{dx}{dt} = -gt$

$dx = -gt\,dt$

$x = -g\frac{t^2}{2} + c \qquad x(0) = h$

$h = c$

$x(t) = -g\frac{t^2}{2} + h$

2. Calcule el valor mínimo de la integral

$$I = \int_0^1 \left[(y')^2 + 12xy\right] dx$$

donde la función $y(x)$ satisface $y(0) = 0$ y $y(1) = 1$.

$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} = 0$

$\mathcal{L} = (y')^2 + 12xy$

$\mathcal{L} = \left(\frac{dy}{dx}\right)^2 + 12xy$

$\frac{\partial \mathcal{L}}{\partial y} = 12x$

$\frac{\partial \mathcal{L}}{\partial y'} = 2y'$

$\frac{d}{dx}\left(\frac{\partial \mathcal{L}}{\partial y'}\right) = 2y''$

$v = 3x^2 + c$

$\frac{dy}{dx} = 3x^2 + c$

$dy = 3x^2 + c\,dx$

$y = x^3 + c_1 x + c_2$

$y(0) = 0 \qquad y(1) = 1$

$0 = c_2 \qquad 1 = 1 + c_1 \qquad c_1 = 0$

$y = x^3$

$\frac{d}{dx}\left(\frac{\partial \mathcal{L}}{\partial y'}\right) - \frac{\partial \mathcal{L}}{\partial y} = 0$

$2y'' - 12x = 0$

$y'' - 6x = 0$

$y'' = 6x$

$v' = 6x$

$\frac{dv}{dx} = 6x$

$dv = 6x\,dx$

$v = 6\frac{x^2}{2} + c$

$y'' = v'$

$y = x^3$

$y' = 3x^2 \quad (y')^2 = 9x^4$

$I = \int_0^1 9x^4 + 12x^4\,dx$

$I = 9\frac{x^5}{5} + 12\frac{x^5}{5}\Big|_0^1$

$I = \frac{9}{5} + \frac{12}{5} = \frac{21}{5}$